

Constructions & Monadic Grammar :

Always taking care of TWO command relations

Basic Working Principles (BWP) (book section or pages in red)

connectedness
as invariant

1 — All analyses by function composition
as the only invariant. (1.2)
(3.2)

application
as derivative
concept (ie. variant)

2 — Application is a special case of
composition where both arguments
are raised before composed. (p. 13)

BWP

continued

- 3 a headed construction is governed by (6.1)
a categorial function capturing
both its syntax and semantics
- 4 a non-headed construction is governed by (p.74)
a categorial relation capturing
both its syntax and semantics
- 5 Every element of grammar must bear two command
relations for that to
work. (1.3)

BWP continued

- 6 - The matrix verb (or predicate) determines the overall clausal structure, including its syntax and semantics. (p1)
- 7 - Subcategorization by the verb is sensitive to configurational and referential structure (p5)
to be able to serve two command relations
- 8 - Analytic structure is completely transparent wrt. grammar and the invariant. (1.3)

BWP continued

- 9 All case is abstract $\equiv$ It is determined by categories as functions. (4.2)
- !! structural / morphological case is a consequence, not cause.
- 10 All case functions are inferrable from verbal subcategorization. (4.2) (App. B)
- 11 Root, stem, finite forms of the verb (ch. 5) require different case functions.
- 12 Because their semantics is different. (5.4)

BWP continued

- 13 grammatical relations are cross-categorial. (ch.8)
- 14 Since categories of verbs are language-specific, cross-linguistic variation is not only possible but expected. (PART II)
- 15 Categorical typology is limited by the invariant (connectedness) (ch.6)
- 16 Accusative, ergative, mixed, split, partial systems are possible & expected. (ch.8)

BWP continued

- 17 All constructions that take a (partial) clause as argument assume that the arguments of the clause are cased (i.e., bear case functions) (4.2)
- 18 Nominal argument-taking is not cased. (11.1)
- 19 That's because case is who **DOES** what to whom. (4.2)
5.2

BWP in closing. In Monadic Grammar:

- 20 - Morphology is the category constructor.
It constructs categories from configuration and reference. (ch.5)
- 21 - Semantics is the constructor of l-command. (2.2)
- 22 - Syntax is transmission of command relations
via the s-command. ANALYSIS IS SYNTAX DRIVEN. (1.4)
- 23 - Phonology is study of transmission medium. (ch.3)

Example construction: Relativization

Some facts to keep in mind before abstracting
for the construction (THE IDEA IS THE SAME)

- Some languages are extracting (English, Turkish)
- Some languages are indexing (Navajo)
- Some languages are resumptive (Arabic, Irish)
- Some languages are both extracting and resumptive (Chinese)

The abstraction is

- The relative construction is mathematically a function:

Rel: $S^{NP} \longrightarrow NP^{NP}$

(only one
argument
is involved)

multiple extraction? (p. 247-248)

Rel: $S^{NP} \xrightarrow{\text{rel}} NP^{NP}$

- S^{NP} can be S/NP or S/NP if it is extracting.
- it can be resumptive S^{NP} if not extracting
- NP^{NP} can be headed rel'n or unheaded.
- rel itself is a function. Can be $\backslash$ or $/$.

English relativisation (N^N rather than NP^{NP})

who(m), which, that :: $(N \setminus N) / (S / NP)$

S^{NP} is S / NP
 NP^{NP} is $N \setminus N$

→

$\lambda P. \lambda Q. \lambda x. P_x \wedge Q_x$

relativised clause (a syntactic argument) modified noun (a syntactic argument) shared argument (NOT a syntactic argument)

N :

$\lambda x ((\lambda x_2. \text{read}' \text{'book'}) x_2) x$

$\wedge$
man x

$\overset{=}{\lambda x. \text{read}' \text{'book'}} x$
 $\wedge$
man x

$N \setminus N : \lambda a \lambda x. ((\lambda x_2. \text{read}' (\text{def}' \text{'book'}) x_2) x) \wedge a x$

$S \setminus NP : \lambda x_2. \text{read}' (\text{def}' \text{'book'}) x_2$

NP
: def' book'

NP/N

: def'

The

N

: man'

man

$(N \setminus N) / (S \setminus NP)$

$\lambda p \lambda a \lambda x. p x \wedge a x$

who

$(S \setminus NP) / NP$

$\lambda x_1 \lambda x_2. \text{read}' x_1 x_2$

read

NP/N

: def'

the

N

: book'

book

N :

$\lambda x. \text{read}' x \bar{J}'$
 $\wedge$
 $\text{hook}' x$

$N \setminus N: \lambda Q \lambda x. ((\lambda x_1. \text{read}' x_1 \bar{J}') x) \wedge Q x$

$S/np: \lambda x_1. \text{read}' x_1 \bar{J}'$

compose

$S/(S \setminus np)$

$: \lambda P. P \bar{J}'$

(case

np/n
 $: \text{def}'$

The

N

$: \text{hook}'$

hook

$(N \setminus N)/(S \setminus np)$

$\lambda P \lambda Q \lambda x. P x \wedge Q x$

that

NP

$: f'$

John

$(S \setminus np)/NP$

$\lambda x_1 \lambda x_2. \text{read}' x_1 x_2$

read

Turkish Relativization

(also extracting)
but rel marker
is morphologically
realized and
syntactically
autonomous)

Rel: $S^{NP} \xrightarrow{\text{rel}} NP^{NP}$

- Two strategies: finite relative clause (-DIK)
non finite relative clause (-yan)

(11.15)

p 237

a. kediyi oksayan misafir

b. * kediyi oksadığı misafir

d. misafirin oksadığı kedi

c. * misafirin oksayan kedi

(Analysis are in the book)

- yan :: (NP/NP) | ^{nonfinite}IV : $\lambda P \lambda Q \lambda x. Px \wedge Qx$

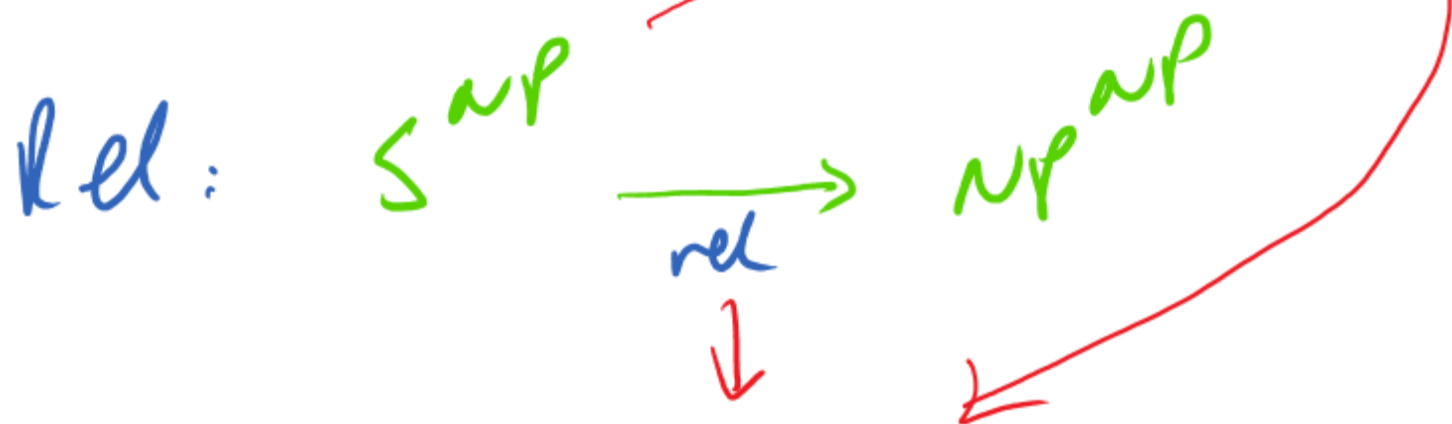
- dik-AGR :: (NP/NP) \ (Sagr \ NP) : $\lambda P \lambda Q \lambda x. Px \wedge Qx$
_{finite}

$\left. \begin{array}{l} \text{SMP is IV or S/NP} \\ \text{NP}^{\text{NP}} \text{ is NP/NP} \end{array} \right\} \text{p. 236 from the book}$
(work out the semantics of analyses at p. 237)

Relativization without extraction: **Navajo**

- Navajo relative marker indexes the arguments in both clauses.

p. 220-221



Abstraction

- e'e :: $(S/S) \setminus S : \lambda P_e \lambda Q. Q \text{ sk}_{\text{topic}(e)}^P \cong \lambda P_e \lambda Q \lambda x. (P_x \wedge Qx)(\text{topic}'_e)$

S^{NP} is realized as S because of non-extraction

NP^{NP} is realized as S/S because they are clausal adjuncts (without extraction)

Rel: $S^{NP} \xrightarrow{\text{rel}} NP^{NP}$

Abstraction

- $e'e \quad \therefore (S/S) \setminus S : \lambda P_e \lambda Q. Q \text{ sk}_{\text{topic}(e)}^P \cong \lambda P_e \lambda Q \lambda x. (P_x \wedge Qx) (\text{topic}_e)$